

ON THE DISTRIBUTION OF A GENERALISED POSITIVE SEMIDEFINITE QUADRATIC FORM OF NORMAL VECTORS

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1. Introduction

Let the elements of $X: p \times n$ be distributed normal with $E(X) = 0$ and the covariance matrix of $\underline{x}'_0 = (\underline{x}'_1, \underline{x}'_2, \dots, \underline{x}'_p)$, where $\underline{x}'_i$ is the i -th row of X , is the Kronecker product $\Sigma_1 \otimes \Sigma_2$ (Σ_1 and Σ_2 are both positive definite). Thus the density function of X is given by:

$$(1.1) \quad f(X) = (2\pi)^{-\frac{1}{2}np} |\Sigma_1|^{-\frac{1}{2}n} |\Sigma_2|^{-\frac{1}{2}p} \text{etr}(-\frac{1}{2}\Sigma_1^{-1}X\Sigma_2^{-1}X')$$

Khatri (1966) derived the distribution of the quadratic form XLX' , for L positive definite. In this paper the distribution of the quadratic form $S = DXAXD'$ for A positive semi-definite of rank r and $D: d \times p$ of rank d is considered. The distribution of S is first derived for the case $r \geq p \geq d$. In part 3 the density of the characteristic roots of S is given as well as the density of the roots in the singular case, that is $d \geq n \geq r$. In part 4 the distribution of $\text{tr } S$ is considered for S non-singular and for S singular. In parts 5 and 6 some special cases are considered and in part 7 the moments of $|S|$ are given. In part 8 a summary of corresponding results for the complex normal distribution are given.

2. A generalized quadratic form

Let the density function of X be given by (1.1). Let $S = DXAX'D'$ where rank $D: d \times p = d$, rank $A: n \times n = r$ and $r \geq p \geq d$. Let $A = A^{\frac{1}{2}'} A^{\frac{1}{2}}$ where rank $A^{\frac{1}{2}'}: r \times n = r$. The density function of $Y = DXA^{\frac{1}{2}'}$ is given by

$$(2.1) \quad g(Y) = (2\pi)^{-\frac{1}{2}dr} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} \text{etr}(-\frac{1}{2}\Sigma_D^{-1}Y\Sigma_A^{-1}Y')$$

where $\Sigma_D = D\Sigma_1 D'$ and $\Sigma_A = A^{\frac{1}{2}}\Sigma_2 A^{\frac{1}{2}'}$

The density function of $S = YY' = DXAX'D'$ is therefore given by

$$(2.2) \quad h(S) = \int_{Y Y' = S} g(Y) dY$$

$$= 2^{-\frac{1}{2}rd} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} \{\Gamma_d(\frac{1}{2}r)\}^{-1} \\ |\Sigma|^{\frac{1}{2}(r-d-1)} {}_0F_0^{(r)}(\Sigma_D^{-1}S, -\frac{1}{2}\Sigma_A^{-1}), S > 0,$$

by using lemma 8 of Khatri (1966).

Interesting to note is that if $A\Sigma_2$ is idempotent, then $\Sigma_A = I_r$ and the hypergeometric function ${}_0F_0$ in (2.2) becomes

$${}_0F_0^{(r)}(\Sigma_D^{-1}S, -\frac{1}{2}I_r) = \sum_{k=0}^{\infty} \sum_K (-\frac{1}{2})^k C_K(\Sigma_D^{-1}S) C_K(I_r)/k! C_K(I_r) \\ = \text{etr}(-\frac{1}{2}\Sigma_D^{-1}S).$$

In this case S is distributed as Wishart with r degrees of freedom.

3. The distribution of the characteristic roots of $S = DXAXD'$

Let X be distributed as in (1.1).

$$HSH' = \Delta = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_d)$$

where $H(dx dx)$ is orthogonal and $\lambda_1 > \lambda_2 \dots \lambda_d$ are the characteristic roots of S . The jacobian of the transformation being

$$\alpha_d(\Delta) = \prod_{i < j} (\lambda_i - \lambda_j).$$

Since (Constantine (1963)):

$$(3.1) \quad \frac{1}{2^d} \int {}_0F_0^{(r)}(\Sigma_D^{-1}H'\Delta H, -\frac{1}{2}\Sigma_A^{-1}) dH \\ = \pi^{\frac{1}{2}d^2} \{\Gamma_d(\frac{1}{2}d)\}^{-1} {}_0F_0^{(r,d)}(\Sigma^{-1}\Delta, -\frac{1}{2}\Sigma^{-1})$$

where ${}_0F_0^{(r,d)}(\Sigma^{-1}\Delta, -\frac{1}{2}\Sigma^{-1})$

$$= \sum_{k=0}^{\infty} \sum_K C_K(\Sigma_D^{-1}) C_K(\Delta) C_K(-\frac{1}{2}\Sigma_A^{-1}) / \{k! C_K(I_d) C_K(I_r)\}$$

the joint density of the characteristic roots of S is given by

$$(3.2) \quad 2^{-\frac{1}{2}dr} \pi^{\frac{1}{2}d^2} \{\Gamma_d(\frac{1}{2}r) \Gamma_d(\frac{1}{2}d)\}^{-1} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} \\ |\Delta|^{\frac{1}{2}(r-d-1)} \alpha_d(\Delta) {}_0F_0^{(d,r)}(\Sigma_D^{-1}\Delta, -\frac{1}{2}\Sigma_A^{-1}), \quad \Delta > 0.$$

Since the characteristic roots of $S = DXAX'D'$ are the same as that of $S^* = A^{1/2}X'D'DXA^{1/2}$, it follows that the density of the non-null characteristic roots of S for $d \geq n \geq r$, i.e. for S singular, is given by

$$(3.3) \quad 2^{-1/2} r d \pi^{-1/2} r^2 \{ \Gamma_r(1/2 d) \Gamma_r(1/2 r) \}^{-1} |\Sigma_D|^{-1/2} |\Sigma_A|^{-1/2} d \cdot \\ \cdot |\Delta|^{1/2(d-r-1)} \alpha_r(\Delta) {}_0F_0(r, d) (\Sigma_D^{-1} \Delta, -1/2 \Sigma_A^{-1})$$

4. The distribution of $\text{tr } S$

The density function of $w = \text{tr } S$ is given by

$$k(w) = \int_{\text{tr}(S)=w} v(\Delta) d\Delta$$

where $v(\Delta)$ is given by (3.2). Let $z_1 = \lambda_1/w$, i.e.

$$z_1 = 1 - z_2 - z_3 - \dots - z_d$$

The jacobian being

$$J(\Delta \rightarrow w, z_2, \dots, z_d) = w^{(d-1)}.$$

$$C_K(\Delta) = w^k C_K(Z)$$

$$\alpha_d(\Delta) = w^{1/2 d(d-1)} \alpha_d(Z)$$

$$|\Delta|^{1/2(r-d-1)} = w^{1/2 d(r-d-1)} |Z|^{1/2(r-d-1)}$$

where $Z = \text{diag}(z_1, z_2, z_3, \dots, z_d)$.

The joint density of $w, z_2, \dots, z_d$ is given by

$$(4.1) \quad 2^{-1/2} r d \pi^{-1/2} d^2 \{ \Gamma_d(1/2 r) \Gamma_d(1/2 d) \}^{-1} |\Sigma_D|^{-1/2} |\Sigma_A|^{-1/2} d \cdot \\ \cdot w^{1/2 r d - 1} |Z|^{1/2(r-d-1)} \alpha_d(Z) \sum_{k=0}^{\infty} \sum_K w^k \\ \cdot C_K(\Sigma_D^{-1}) C_K(Z) C_K(-1/2 \Sigma_A^{-1}) / \{k! C_K(I_d) C_K(I_r)\}$$

From Khatri and Pillai (1968) we have that

$$(4.2) \quad \int_D |Z|^{1/2(r-d-1)} \alpha_d(Z) C_K(Z) dz_2 \dots dz_d \\ = \Gamma_d(1/2 r) \Gamma_d(1/2 d) {}_d(1/2 r)_K C_K(I_d) / \{ \pi^{1/2} d^2 \Gamma(1/2 r d) {}_d(1/2 r d)_K \}$$

where $D = \{0 < z_d < \dots < z_2 < z_1 = 1 - z_2 \dots z_d\}$

and $d(a)_K = \Gamma_d(a, K) / \Gamma_d(a)$.

Using this integral it follows from (4.1) that the density of w is

$$(4.3) \quad 2^{-\frac{1}{2}rd} \{\Gamma(\frac{1}{2}rd)\}^{-1} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} d_w^{\frac{1}{2}rd-1} \\ \cdot \sum_{k=0}^{\infty} \sum_K d(\frac{1}{2}r)_K (-\frac{1}{2}w)^k C_K(\Sigma_D^{-1}) C_K(\Sigma_A^{-1}) / \{k! (\frac{1}{2}rd)_k C_K(I_r)\}$$

for $w > 0$.

If $d \geq n \geq r$, i.e. S singular, the density is given by

$$(4.4) \quad 2^{-\frac{1}{2}rd} \{\Gamma(\frac{1}{2}rd)\}^{-1} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} d_w^{\frac{1}{2}rd-1} \\ \cdot \sum_{k=0}^{\infty} \sum_K r(\frac{1}{2}d)_K (-\frac{1}{2}w)^k C_K(\Sigma_D^{-1}) C_K(\Sigma_A^{-1}) / \{k! (\frac{1}{2}rd)_k C_K(I_d)\}$$

5. A quadratic form in singular normal variates

Let the rows of $X:pxn$ be independently distributed $N(\underline{0}, B)$ where $\text{rank } BA = t \geq p$ for a positive semi-definite matrix A .

Let

$$X = YL$$

such that the rows of $Y(p \times q)$ are independently distributed $N(\underline{0}, I_q)$, $q = \text{rank } L = \text{rank } B$ and $B = L'L$. It follows that

$$S = XAX' = YLAL'Y'$$

with $\text{rank } LAL' = \text{rank } BA = t \leq \min(q, r)$.

The density of S now follows from (2.2) as:

$$(5.1) \quad 2^{-\frac{1}{2}pt} \{\Gamma_p(\frac{1}{2}t)\}^{-1} |\Delta_{AB}|^{-\frac{1}{2}p} |S|^{\frac{1}{2}(t-p-1)} {}_0F_0^{(t)}(S, -\frac{1}{2}\Delta_{AB}^{-1}), \quad S > 0$$

where $\Delta_{AB}(\text{txt})$ is a diagonal matrix with diagonal elements the non null characteristic roots of AB .

The density of the characteristic roots of S follows from (3.2) as:

$$(5.2) \quad 2^{-\frac{1}{2}pt} \pi^{\frac{1}{2}p^2} \{\Gamma_p(\frac{1}{2}t) \Gamma_p(\frac{1}{2}p)\}^{-1} |\Delta_{AB}|^{-\frac{1}{2}p} |\Delta|^{-\frac{1}{2}(t-p-1)} \\ \cdot \alpha_p(\Delta) {}_0F_0^{(t)}(\Delta, -\frac{1}{2}\Delta_{AB}^{-1}), \quad \Delta > 0$$

and the density of $w = \text{tr } S$ from (4.3) as:

$$(5.3) \quad 2^{-\frac{1}{2}p} |\Gamma(\frac{1}{2}tp)|^{-1} |\Delta_{AB}|^{-\frac{1}{2}} p_w^{\frac{1}{2}tp-1} \\ \cdot \sum_k \sum_K p^{\frac{1}{2}t} K^{(-\frac{1}{2}w)^k} C_K(I_p) C_K(\Delta_{AB}^{-1}) / \{(\frac{1}{2}tp)_k \\ \cdot k! C_K(I_t)\}.$$

6. Special cases

(a) The singular Wishart

Let the columns of $X: p \times n$ be independently distributed $N(\underline{0}, \Sigma)$ with $n \leq p$. The density of X is given by:

$$(2\pi)^{-\frac{1}{2}np} |\Sigma|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Sigma^{-1}XX').$$

The joint density of the roots of $S = XX'$ follows from (3.3) as:

$$(6.1) \quad 2^{-\frac{1}{2}np} \pi^{\frac{1}{2}n^2} (\Gamma_n(\frac{1}{2}p)\Gamma_n(\frac{1}{2}n))^{-1} |\Sigma|^{-\frac{1}{2}n} |\Delta|^{\frac{1}{2}(p-n-1)} \\ \alpha_n(\Delta) {}_0F_0^{(n)}(-\frac{1}{2}\Sigma^{-1}, \Delta), \Delta > 0.$$

The density of $w = \text{tr } S$ is given by

$$(6.2) \quad 2^{-\frac{1}{2}np} \{\Gamma(\frac{1}{2}np)\}^{-1} |\Sigma|^{-\frac{1}{2}n} w^{\frac{1}{2}np-1} \\ \cdot \sum_k \sum_n (\frac{1}{2}p)_K^{(-\frac{1}{2}w)^k} C_K(\Sigma^{-1}) C_K(I_n) / \{k! (\frac{1}{2}np)_k C_K(I_p)\} \quad \text{for } w > 0.$$

James (1960) derived the joint density of the characteristic roots of S for $n \geq p$.

(b) If $n = 1$

An interesting special case is when only one sample is drawn i.e. $n = 1$.

Since XX' is then of rank one with only one non-zero root, say λ , it follows from (6.1) that the density of λ is given by

$$(6.3) \quad 2^{-\frac{1}{2}p} \{\Gamma(\frac{1}{2}p)\}^{-1} |\Sigma|^{-\frac{1}{2}} \lambda^{\frac{1}{2}p-1} {}_0F_0^{(p)}(-\frac{1}{2}\Sigma^{-1}, \lambda), \quad \lambda > 0$$

which is also given by James (1964). By choosing Σ equal to the following special cases we obtain certain interesting densities.

(i) Let $\Sigma = \text{diag}(\sigma_1^2, \sigma_1^2, \dots, \sigma_p^2)$

($\sigma_i^2, i=1, 2, \dots, p$, can be chosen as the characteristic roots of Σ).

Then

$$\begin{aligned}\lambda &= \mathbf{X}'\mathbf{X} \\ &= \sum_{i=1}^p \mathbf{X}_i^2\end{aligned}$$

where $\mathbf{X}_i$ is $N(0, \sigma_i^2)$ distributed, independently of $\mathbf{X}_j$, $i \neq j$, $i = 1, 2, \dots, p$. The density is given by (6.3).

(ii) Let $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_1^2, \dots, \sigma_q^2, \dots, \sigma_q^2)$ i.e. p is partitioned into q parts, m_1 of the first, m_2 of the second, etc., such that

$$\sum_{i=1}^q m_i = m.$$

Then

$$\begin{aligned}\lambda &= \mathbf{X}'\mathbf{X} \\ &= \sum_{i=1}^q \mathbf{X}_i^2\end{aligned}$$

where $\mathbf{X}_i^2$ is distributed independently according to $\sigma_i^2 \chi^2(m_i)$.

The density follows from (6.3) as

$$(6.4) \quad 2^{-\frac{1}{2}m} \{\Gamma(\frac{1}{2}m)\}^{-1} |\Sigma|^{-\frac{1}{2}} \lambda^{\frac{1}{2}m-1} {}_0F_0^{(m)}(-\frac{1}{2}\Sigma^{-1}, \lambda), \quad \lambda > 0.$$

(iii) Let $\Sigma = \text{diag}(\Sigma_1, \dots, \Sigma_q)$. Then

$$\lambda = \mathbf{X}'\mathbf{X} \text{ where } \mathbf{X}' = (\mathbf{X}^{(1)'}, \mathbf{X}^{(2)'}, \dots, \mathbf{X}^{(q)'})$$

and $\mathbf{X}^{(i)}$ distributed independently according to $N(0, \Sigma_i)$.

The density follows from (6.3) as

$$(6.5) \quad \frac{2^{-\frac{1}{2}pq}}{\Gamma(\frac{1}{2}pq)} |\Sigma|^{-\frac{1}{2}} \lambda^{\frac{1}{2}pq-1} {}_0F_0^{(pq)}(-\frac{1}{2}\Sigma^{-1}, \lambda), \quad \lambda > 0.$$

For the above case we can also consider

$$\lambda = \sum_{i=1}^q \mathbf{X}_i' \mathbf{X}_i$$

where $\mathbf{X}_i(p_i \times 1)$ is distributed independently $N(0, \Sigma_i)$. The density of λ is given by

$$(6.6) \quad \frac{2^{-\frac{1}{2}P} |\Sigma|^{-\frac{1}{2}}}{\Gamma_q(\frac{1}{2}P)} \lambda^{\frac{1}{2}P-1} {}_0F_0^{(P)}(\lambda, -\frac{1}{2}\Sigma^{-1})$$

where $P = \sum_{i=1}^q p_i$.

(iv) Let $\Sigma = \text{diag}(\Sigma_1, \dots, \Sigma_1, \dots, \Sigma_q, \dots, \Sigma_q)$ i.e. Σ consists of m matrices in the diagonal, m_1 of the first kind, m_2 of the second, etc.

Then

$$\lambda = X'X$$

where

$$X' = (X^{(1,1)'}, X^{(1,2)'}, \dots, X^{(1,m_1)'}, \dots, X^{(q,1)'}, \dots, X^{(q,m_q)'})$$

and

$X^{(ij)}$ is distributed independently according to $N(0, \Sigma_j)$ $j = 1, 2, \dots, m_j$. The density follows as

$$(6.7) \quad 2^{-\frac{1}{2}mp} \{\Gamma(\frac{1}{2}mp)\}^{-1} \prod_{i=1}^q |\Sigma_i|^{-\frac{1}{2}m_i} \lambda^{\frac{1}{2}mp-1} {}_0F_0^{(mp)}(-\frac{1}{2}\Sigma^{-1}, \lambda), \quad \lambda > 0.$$

7. On the moments of $|S|$

If the density of $X: p \times n$ is given by (1.1), then the density of $S = DXAX'D'$, given by (2.2), can also be written in the form:

$$(7.1) \quad h(S) = 2^{-\frac{1}{2}rd} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^{-\frac{1}{2}d} \{\Gamma_d(\frac{1}{2}r)\}^{-1} |S|^{\frac{1}{2}(r-d-1)} \cdot \text{etr}(-\frac{1}{2}\Sigma_D^{-1}S) {}_0F_0^{(r)}(\Sigma_D^{-1}S, \frac{1}{2}(I - \Sigma_A^{-1})), \quad S > 0.$$

Since (Constantine (1963)):

$$(7.2) \quad \int_{S>0} \text{etr}(-\frac{1}{2}\Sigma_D^{-1}S) |S|^{\frac{1}{2}(r-d-1)+h} {}_0F_0^{(r)}(\Sigma_D^{-1}S, \frac{1}{2}(I - \Sigma_A^{-1})) dS \\ = \Gamma_d(\frac{1}{2}r+h) 2^{\frac{1}{2}r+h} \Gamma_d(\frac{1}{2}r+h; I_d, \frac{1}{2}(I - \Sigma_A^{-1}))$$

it follows that the h -th moment of $|S|$ is given by:

$$(7.3) \quad E(|S|^h) = 2^{dh} \Gamma_d(\frac{1}{2}r+h) \{\Gamma_d(\frac{1}{2}r)\}^{-1} |\Sigma_D|^h |\Sigma_A|^{-\frac{1}{2}d} \\ \cdot {}_1F_0^{(r)}(\frac{1}{2}r+h; I_d, \frac{1}{2}(I - \Sigma_A^{-1}))$$

If $d \geq n \geq r$, i.e. if S is singular, let

$\Delta = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$ are the non-null characteristic roots of S . Since

$$|\Delta| = |A^{\frac{1}{2}} X' D' D X A^{\frac{1}{2}}|, \text{ it follows that:}$$

$$(7.4) \quad E(|\Delta|^h) = 2^{rh} \Gamma_r(\frac{1}{2}d+h) \{\Gamma_r(\frac{1}{2}d)\}^{-1} |\Sigma_D|^{-\frac{1}{2}r} |\Sigma_A|^h \\ \cdot {}_1F_0^{(d)}(\frac{1}{2}d+h; I_r, \frac{1}{2}(I - \Sigma_D^{-1})).$$

8. Summary of corresponding results in the case of complex normal vectors

Let the density function of the complex matrix $X: p \times n$ be given by:

$$(8.1) \quad f(X) = \pi^{-np} |\Sigma_1|^{-n} |\Sigma_2|^{-p} \text{etr}(-\Sigma_1^{-1} X \Sigma_2^{-1} X')$$

Let $S = D X A \bar{X}' D'$, where $\text{rank } D: d \times p = d$, $\text{rank } A: n \times n = r$, A being hermitian and positive semi-definite and $r \geq p \geq d$. The density function of S is given by:

$$(8.2) \quad h(S) = \{\tilde{\Gamma}_d(r)\}^{-1} |\Sigma_D|^{-r} |\Sigma_A|^{-d} |S|^{r-d} {}_0\tilde{F}_0^{(r)}(\Sigma_D^{-1} S, -\Sigma_A^{-1}), \quad S > 0$$

where $\Sigma_D = D \Sigma_1 \bar{D}'$, $\Sigma_A = A^{\frac{1}{2}} \Sigma_2 \bar{A}^{\frac{1}{2}}$, and the complex multivariate gamma function, $\tilde{\Gamma}_d(r)$, and the complex hypergeometric function, is defined by James (1964). If $A \Sigma_2$ is idempotent, then it is obvious that S has a complex Wishart distribution.

The density of the (real) characteristic roots of S is given by:

$$(8.3) \quad \pi d(d-1) \{\tilde{\Gamma}_d(r) \tilde{\Gamma}_d(d)\}^{-1} |\Sigma_D|^{-r} |\Sigma_A|^{-d} |\Delta|^{r-d} \tilde{\alpha}_d(\Delta) \\ \cdot {}_0\tilde{F}_0^{(r,d)}(\Sigma_D^{-1} \Delta, -\Sigma_A^{-1}), \quad \Delta > 0$$

where $\Delta = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $\lambda_1 > \dots > \lambda_d$ are the non-null characteristic roots of S ,

$$\tilde{\alpha}_d(\Delta) = \prod_{i < j} (\lambda_i - \lambda_j)^2 \text{ and}$$

$${}_0\tilde{F}_0^{(r,d)}(\Sigma_D^{-1} \Delta, -\Sigma_A^{-1}) = \sum_{k=0}^{\infty} \sum_k \tilde{C}_k(\Sigma_D^{-1}) \tilde{C}_k(\Delta) \tilde{C}_k(-\Sigma_A^{-1}) / \{k! \tilde{C}_k(I_d) \tilde{C}_k(I_r)\}$$

If $d \geq n \geq r$, i.e. if S is singular, the density of the non-null characteristic roots of S is given by:

$$(8.4) \quad \pi^{r(r-1)} \{\tilde{\Gamma}_r(d) \tilde{\Gamma}_r(r)\}^{-1} |\Sigma_D|^{-r} |\Sigma_A|^{-d} |\Delta|^{d-r} \\ \cdot \tilde{\alpha}_r(\Delta) {}_0\tilde{F}_0(r, d) (\Sigma_D^{-1} \Delta, -\Sigma_A^{-1}), \quad \Delta > 0$$

The h -th moment of $|S|$ is given by:

$$(8.5) \quad E(|S|^h) = \tilde{\Gamma}_d(r+h) \{\tilde{\Gamma}_d(r)\}^{-1} |\Sigma_D|^h |\Sigma_A|^{-d} {}_1\tilde{F}_0(r) (r+h; I_d, I - \Sigma_A^{-1})$$

and if $d \geq n \geq r$, then:

$$(8.6) \quad E(|\Delta|^h) = \tilde{\Gamma}_r(d+h) \{\tilde{\Gamma}_r(d)\}^{-1} |\Sigma_D|^{-r} |\Sigma_A|^h {}_1\tilde{F}_0(d) (d+h; I_r, I - \Sigma_D^{-1})$$

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